Refactoring Plan: Biostatistics Practicum

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Purpose and method

This document proposes a refactoring of *Biostatistics Practicum* against the evidence base assembled for the blog post 'What Makes an Open Textbook Work' (`~/prj/ryyblog/posts/pub-open-textbook-guidebook/`). The goal stated for the exercise was engagement and impact. Those are not directly measurable from the manuscript, so this plan uses the proxies that the literature actually supports: whether students can practice with feedback, whether the book is navigable and assignable in pieces, whether it is legally and practically adoptable by other instructors, and whether it presents anything to look at.

The findings below rest on a mechanical scan of all 38 content files in `analysis/report/` performed on 2026-09-07, not on a reading of the prose. Counts are reproducible from the scripts noted in the appendix. Where a number is an estimate rather than a count, it says so.

Two prior documents overlap with this one and are not superseded. `DEEP-REVIEW.md` (2026-07-05) covered voice, disciplinary framing, and peer comparison. `REVIEW-CHECKLIST.md` (2026-09-07) is the per-chapter ownership review, currently open at chapter `00-intro`. This plan is concerned with structure and adoptability, and deliberately says nothing about prose quality, which the rgtvoice pass has already addressed.

Executive summary

The manuscript is in better condition than most open textbooks on the criteria reviewers rate most often, and is in poor condition on three criteria that the evidence says matter most for learning and adoption.

The book has roughly 180 exercises and no solutions to any of them. It has 226 display-only code blocks against 48 executable ones, so the large majority of its code cannot be verified by rendering and produces no output. Thirty of its 38 chapters contain exactly one figure, and that figure is a schematic diagram rather than a computed result. It carries a GPL-3 license, which is a software license and will not be accepted by the open-textbook catalogs through which adoption actually happens. It has no glossary and no index, and no alt text anywhere.

None of these is a writing problem. All five are production decisions that can be reversed in a defined amount of work, and the first three account for most of the available gain.

Part 1. Measured state

Property	Measured	Target from evidence
Total words	111,865	n/a
Estimated pages (at 450 w/p)	~248	Downey: ~140 per semester course
Chapters and appendices	38	n/a
Median chapter length	~6 pages	one sitting; acceptable as is
Chapters over 9 pages	5	split or justify
Exercises (approx.)	~180	n/a
Exercises with published solutions	0	OpenIntro: nearly all
Chapters with exactly 5 exercises	30 of 38	proportional to chapter weight
Executable `{r}` chunks	48	code that renders is code that is checked
Display-only fenced blocks	226	n/a
Computed figures (R-generated)	~20, in 8 chapters	reviewers flag words-only texts
Chapters whose only figure is a mermaid diagram	30	n/a
Chapters with zero figures	4	zero
Files containing `fig-alt`	0 of 46	all informative figures
Glossary	absent	present
Index	absent	present
License	GPL-3	CC BY
Output formats	HTML, PDF	plus print-on-demand
`freeze: auto`	set	correct as is
Bibliography entries	99, recency good	correct as is
'Check your understanding' sections	1 per chapter	1 per major section

Two things in that table are worth stating plainly as strengths, because they are unusual. The bibliography is current, with 25 entries from 2024 and several from 2026, so the staleness signal that drives de-adoption is absent. And the book already has a per-chapter prerequisites quiz with published answers, which is a retrieval practice structure most textbooks lack entirely. The refactoring extends that structure rather than inventing one.

Part 2. Findings, ranked by expected impact

Finding 1. About 180 exercises, zero solutions

Measured. Every chapter except `00-intro`, `25-synthesis`, and the three appendices carries an `## Exercises` section, almost always with five numbered items. A search for solution or answer headings returns only the per-chapter `Prerequisites answers` sections, which answer the opening quiz and not the exercises.

Why it ranks first. Dunlosky and colleagues rate practice testing as one of only two high-utility study techniques, and the mechanism depends on feedback: a student who cannot check an answer is not doing retrieval practice, only doing homework. OpenIntro ships more than 550 exercises with solutions to nearly all in-text exercises, placed in footnotes on the explicit principle that the check should be immediate. Reviewers in the Open Textbook Library corpus complain when solutions are merely at the end of the chapter; this book does not have them at all. For a self-studying reader, which an open book acquires in quantity, an exercise without an answer is close to worthless.

Complication specific to this book. Roughly half the exercises instruct the student to work on their own project or their own data, which admits no answer key. That is a defensible design for a practicum, but it cannot be the whole exercise set, because it means a reader without an ongoing project cannot self-assess anywhere in the book.

Action. Split the exercise set in two. Retain the open-ended 'apply to your own work' items, and mark them as such so the reader knows no key exists. Add, to every chapter, three to five exercises that operate on a dataset the book ships, and publish worked solutions for those. Place solutions where OpenIntro places them, which is close to the question, either in a collapsible callout immediately following the exercise or in a per-chapter solutions section reached by anchor link rather than by scrolling.

Finding 2. 226 display-only code blocks against 48 executable

Measured. 226 fenced blocks marked as plain `r`, `bash`, or `sh` against 48 executable `{r}` chunks, with only one chunk explicitly marked `eval: false`. The display-only blocks are not chunks Quarto declined to run; they are blocks that were never wired to run.

Why it matters. Two consequences follow, and the second is the expensive one. First, none of that code is verified by rendering, so a typo in a display-only block survives every build indefinitely, and `freeze: auto` does not help because there is nothing to freeze. Second, code that does not execute produces no output, which is the direct upstream cause of Finding 3. A chapter on plotting that shows plotting code without plots is the canonical reviewer complaint in this literature.

Realistic scope. Much of this code genuinely cannot execute at render: shell sessions, Docker builds, `git` interactions, cloud provisioning, and SAS. That is a legitimate reason for a display block, and those should stay display-only and be verified another way. The target is the subset that could execute and simply is not: the wrangling, types, joins, graphics, missing-data, CDISC, and case-study chapters.

Action. Triage all 226 blocks into three categories. Executable and should run, convert to `{r}` chunks. Not executable in principle, mark explicitly and add a verification note saying how the command was checked and when. Executable but expensive, run once and cache with `freeze`. Do the conversion chapter by chapter, verifying the render after each, since converting a block that errors will break the build.

Finding 3. Thirty chapters whose only figure is a schematic

Measured. Only eight chapters contain any R-generated figure: `11-quarto` (3), `12-rmd-workflow` (2), `13-wrangling` (1), `16-graphics` (5), `17-missing` (2), `19-cdisc` (1), `23-penguins` (4), and `25d-ethics` (2), for roughly 20 computed figures in total. Thirty chapters contain exactly one figure, a mermaid diagram. Four files contain none at all, including `25b-communicating`, which at about 11 estimated pages is among the longest in the book.

Why it matters. 'The textbook contains only words' is a verbatim reviewer complaint from the Open Textbook Library corpus, offered with the prediction that it costs student engagement. The mermaid diagrams are genuinely useful and should stay; the point is that a schematic of a workflow is not the same as seeing what the code produced.

Relationship to `DEEP-REVIEW.md`. That document identified this exact defect in July 2026 and reported it addressed 'in the highest-value chapters'. The measurement confirms the fix was real and partial: the eight chapters with computed figures are presumably those chapters. This plan proposes finishing the job.

Action. Set a floor of one computed figure or rendered table per major technical section, not per chapter. Priority order is the chapters where a visual is load-bearing and currently absent: `25b-communicating` (zero figures in a chapter about communicating results is the sharpest single instance), `14-types`, `15-joins`, `15b-databases`, `12b-acquisition`, `24-adni`, `17-missing` beyond its current two, and `20-testing`. This work is largely a byproduct of Finding 2 and should be scheduled with it.

Finding 4. GPL-3 license blocks catalog adoption

Measured. `LICENSE.md` and `DESCRIPTION` both declare GPL-3.

Why it matters. GPL-3 is a software license. Applied to prose it is at best ambiguous about attribution and derivative works in the way a teaching text needs, and more practically it is not a license that the open-textbook catalogs accept. Those catalogs are how adoption happens: faculty report discoverability as a barrier roughly half the time, and the Open Textbook Library's review program is explicitly dual-purpose, supplying both credibility and awareness. A book that cannot be listed forgoes the main distribution channel available to it.

Action. Relicense the prose under CC BY, keeping GPL-3 or MIT for the code in `R/` and the scripts, and state the split explicitly in `LICENSE.md` and in the colophon. Avoid ND and NC variants, which are widely held to disqualify a work as an open educational resource because they forbid the revision and remix that define the category. Confirm that every contributor consents before relicensing, since this cannot be undone unilaterally later.

Finding 5. No glossary, no index, no alt text

Measured. No glossary file or heading anywhere in the book. No index. Zero of 46 files contain a `fig-alt` attribute.

Why it matters. 'There is no index, glossary or table of contents' is, again, a verbatim reviewer complaint, and comprehensiveness in the Open Textbook Library rubric explicitly asks for an effective index or glossary. The book does have a navigable sidebar table of contents, so this is two thirds of a known complaint rather than all of it. Alt text is a WCAG requirement for informative images and an accessibility criterion reviewers apply; at zero coverage the book currently fails it outright.

Action. Three separable tasks. Add a glossary appendix collecting the terms already bolded on first use across the chapters, which makes this largely an extraction job rather than a writing job. Add a back-of-book index for the PDF output. Add `fig-alt` to every figure, which is roughly 50 short descriptions once Findings 2 and 3 have increased the figure count.

Finding 6. Exercise count is uniform, not proportional

Measured. Thirty of 38 chapters carry exactly five exercises, irrespective of chapter weight. `10b-targets` at about 4 estimated pages has five; `12b-acquisition` at about 11 has five.

Why it matters. This is the structural item flagged in `REVIEW-CHECKLIST.md` Phase 1 as 'exercise placement and count proportionate to the chapter's weight'. Uniformity of exactly five is the signature of a template applied evenly rather than a judgment made per chapter, and it means the longest and most demanding chapters are the least practiced.

Action. Rebalance to roughly one exercise per major section, which would put the short chapters near three and the long ones near eight. Fold this into the Finding 1 work rather than doing it separately.

Finding 7. Retrieval density is one check per six pages

Measured. Almost every chapter has exactly one 'Check your understanding' section; `03-team-science` has two.

Why it matters. The existing structure is right and there is simply not much of it. Given the reading-compliance evidence, that most students do not read a chapter straight through, a single mid-chapter check is easy to miss entirely.

Action. Target one short retrieval prompt per major section, with the answer immediately available. This is cheap relative to its evidential support and should be done alongside Finding 1.

Finding 8. Length and modularity

Measured. About 248 estimated pages. Five chapters exceed nine estimated pages: `12b-acquisition` and `25b-communicating` at about 11, `05b-git-teams` and `25d-ethics` at about 10, `22-sas` at about 9.

Why it matters. Downey's target of roughly 140 pages for a semester course is a guideline from one author rather than an empirical result, and a practicum spanning Git through CDISC has a defensible reason to be longer. I would not restructure the book to hit 140. The per-chapter figure matters more, and at a median of about six pages the book is already close to the one-sitting target. Modularity is the weakest criterion across the Open Textbook Library corpus, at 62.5% positive, and this book is better placed than most because every chapter opens with a prerequisites quiz that states what it assumes.

Action. Split only the five long chapters, at natural seams, and leave the total length alone. Make the modularity that already exists explicit by stating in the preface that chapters are assignable individually, since an instructor assigning three chapters of a free book incurs no cost justification.

Finding 9. Adoption mechanics are entirely unaddressed

Measured. The book renders to HTML and PDF and is hosted. There is no evidence in the repository of a catalog listing, an ISBN, a print-on-demand edition, or a list of adopting courses.

Why it matters. Perceived quality is mediated by colleague recommendation, personal knowledge of the author, and peer review rather than by intrinsic merit, and a commercial publisher's marketing resources are, in OpenIntro's phrase, humbling. A book with no distribution strategy is read by the author's own students and nobody else. Separately, more than a quarter of students in one e-textbook pilot believed they would have learned more from paper, and print-on-demand is nearly free to offer.

Action. After the license change, list in the Open Textbook Library and solicit reviews, which is simultaneously a credibility and an awareness act. Produce a print-on-demand paperback from the existing PDF. Publish a page of adopting courses as it accumulates.

Part 3. Phased plan

Effort figures are estimates in working days for one person, offered for sequencing rather than for scheduling, and they assume AI assistance for the mechanical parts with human verification of every result.

Phase A. Adoptability unblock (about 2 days)

Small, self-contained, and a precondition for everything in Phase D.

1. Relicense prose to CC BY, keep code under GPL-3 or MIT, document the split in `LICENSE.md`, `DESCRIPTION`, and the colophon.
2. Fix the two open items from `REVIEW-CHECKLIST.md` on `00-intro`: the appendix count and the missing 'Sources' entry in the chapter pattern list.
3. Add a `fig-alt` to every figure that currently exists.

Acceptance. The license split is stated in three places; `00-intro` describes three appendices; `grep -L fig-alt` returns no file that contains a figure.

Phase B. Practice with feedback (about 8 to 12 days)

The highest-value phase. Do it chapter by chapter in curriculum order.

1. Per chapter, classify existing exercises as open-ended or checkable, and label the open-ended ones explicitly.
2. Per chapter, add three to five checkable exercises built on a dataset the book already ships.
3. Write worked solutions for every checkable exercise and place them adjacent to the question, in a collapsible callout for HTML and a per-chapter solutions block for PDF.
4. Rebalance exercise counts to roughly one per major section.
5. Add a retrieval prompt per major section, with the answer immediately available.

Acceptance. Every chapter has at least three exercises with published solutions; no chapter has exactly five exercises by coincidence of template; a reader with no project of their own can self-assess in every chapter.

Phase C. Executable code and figures (about 10 to 15 days)

1. Triage all 226 display-only blocks into executable, not executable, and expensive.
2. Convert the executable subset to `{r}` chunks, one chapter at a time, rendering after each chapter.
3. For blocks that cannot execute, add a dated verification note stating how and when the command was checked.

4. Bring every major technical section to at least one computed figure or rendered table, prioritizing `25b-communicating`, `14-types`, `15-joins`, `15b-databases`, `12b-acquisition`, `24-adni`, and `20-testing`.
5. Add `fig-alt` to each new figure as it is created.

Acceptance. A full render completes clean; no chapter has zero figures; the count of computed figures is at least one per major technical section.

Phase D. Reference apparatus and distribution (about 5 days)

1. Build the glossary appendix by extracting bolded first-use terms.
2. Add a PDF index.
3. Split the five long chapters at natural seams.
4. State the modular-assignment policy in the preface.
5. List in the Open Textbook Library and solicit reviews.
6. Produce a print-on-demand paperback.

Acceptance. Glossary and index present in both output formats; no chapter over about nine estimated pages; catalog listing submitted.

Part 4. What not to do

- Do not rewrite prose for voice. The rgtvoice pass is complete and `DEEP-REVIEW.md` found the register already consistent across all chapters. Further prose churn would consume the budget that Phase B needs.
- Do not restructure the book to reach 140 pages. The length is defensible for the scope, and the per-chapter length is already close to target.
- Do not delete the mermaid diagrams in favor of computed figures. They serve a different purpose and reviewers do not complain about schematics; they complain about their absence of everything else.
- Do not adopt an NC or ND license variant to retain commercial control. It would disqualify the book from the catalogs that Phase D depends on, and Downey's observation applies: as copyright holder you retain the ability to grant separate licenses later regardless of the public default.
- Do not add humor or decorative photographs to chase engagement. The seductive-details literature suggests entertaining tangents can reduce learning, and OpenIntro's stated practice is to pursue attention through interesting data and questions instead.

Part 5. Sequencing

Phase A first, because it is short and unblocks Phase D. Phase B next, because it carries the most evidential support per unit of work and is independent of Phase C. Phase C next, and it will generate most of the figures Finding 3 asks for as a byproduct. Phase D last, because listing a book in a catalog before its exercises have solutions invites the review one would least like to receive.

If only one phase is ever done, do Phase B.

Part 6. Limitations of this analysis

This is a mechanical scan, not a reading. Word counts include code blocks and YAML, so page estimates run high; the 450-words-per-page divisor is conventional rather than measured against this book's rendered output. Exercise counts come from counting numbered list items under an `## Exercises` heading and will miss exercises formatted differently. The figure counts distinguish mermaid diagrams from R-generated figures by chunk syntax and would misclassify a figure included by another route, though no markdown image includes were found in any chapter.

I have not assessed cultural relevance, which is the second-weakest criterion across the Open Textbook Library corpus at 66.6% positive and which cannot be measured by grep. That requires a reading pass over the examples, and it should be added to the per-chapter review in `REVIEW-CHECKLIST.md` rather than guessed at here.

Nor have I verified that the proposed exercise solutions are feasible for every chapter. Some topics, cloud provisioning and team Git among them, may not admit a self-checkable exercise on a shipped dataset, and those chapters may have to remain open-ended with the limitation stated to the reader.

Finally, the effort estimates are estimates. No part of this plan has been executed, and nothing in it has been tested against the actual difficulty of converting the first display block or writing the first solution set.

Appendix. Reproducing the measurements

The two scan scripts used are in the session scratchpad: `measure.sh` (per-chapter words, exercises, figures, chunks, retrieval sections) and `measure2.sh` (chunk types, alt text, formats, license, exercise style). They are read-only over `analysis/report/` and can be rerun after each phase to confirm the acceptance criteria above.

Rendered on 2026-09-07 at 14:04 PDT. Source: ~/prj/tch/02-practicum/REFACTOR-PLAN.md